

QUESTION 1

SOURCE CODE:

```
#include <iostream>
using namespace std;
class student{
    private:
        string name;
        int roll_no;
        int marks;
    public:
        student(){
            name = "Default";
            roll_no = 101;
            marks = 50;
        }

        student(string n, int r, int m){
            name = n;
            roll_no = r;
            marks = m;
        }
}
```

```
void displayStudentInfo(){
    cout<<"NAME: "<<name<<endl;
    cout<<"ROLL NUMBER: "<<roll_no<<endl;
    cout<<"MARKS: "<<marks<<endl;
}

};

int main(){
    student s1;
    student s2("Areej", 202, 89);
    student s3("Waleeja", 302, 37);
    student s4;

    cout<<"====STUDENT 1 INFO===="<<endl;
    s1.displayStudentInfo();
    cout<<"====STUDENT 2 INFO===="<<endl;
    s2.displayStudentInfo();
    cout<<"====STUDENT 3 INFO===="<<endl;
    s3.displayStudentInfo();
    cout<<"====STUDENT 4 INFO===="<<endl;
    s4.displayStudentInfo();
    return 0;
}
```

OUTPUT:

```
PS C:\Sem 2\OOP LABS\lab7> cd C:\Sem 2\OOP LABS\lab7
====STUDENT 1 INFO====
NAME: Default
ROLL NUMBER: 101
MARKS: 50
====STUDENT 2 INFO====
NAME: Areej
ROLL NUMBER: 202
MARKS: 89
====STUDENT 3 INFO====
NAME: Waleeja
ROLL NUMBER: 302
MARKS: 37
====STUDENT 4 INFO====
NAME: Default
ROLL NUMBER: 101
MARKS: 50
PS C:\Sem 2\OOP LABS\lab7> |
```

QUESTION 2

SOURCE CODE:

```
#include <iostream>
using namespace std;
class result{
    private:
        string student_name;
        int roll_no;
        int marks[5];

    public:
        result(){
            student_name = "Default";
            roll_no = 200;
            for(int i=0;i<5;i++){
                marks[i]=0;
            }
        }
        result(string n, int r, int m[5]){
            student_name = n;
            roll_no = r;
            for(int i=0;i<5;i++){
                marks[i]=m[i];
            }
        }
}
```

```

float calculateAverage(){
    float average;
    int sum=0;
    for(int i=0;i<5;i++){
        sum+=marks[i];
    }
    average = sum/5.0;
    return average;
}

void displayResult(){
    cout<<"NAME: "<<student_name<<endl;
    cout<<"ROLL NUMBER: "<<roll_no<<endl;
    for(int i=0;i<5;i++){
        cout<<"MARKS "<<i+1<<" "<<marks[i]<<endl;
    }
    cout<<"AVERAGE: "<<calculateAverage()<<endl;
}

};

```

```

int main(){
    result R1;
    int marks[5] = {34, 56, 67, 78, 90};
    result R2("waleeja", 101, marks);

    R1.displayResult();
    R2.displayResult();
}

```

OUTPUT:

```
NAME: Default
ROLL NUMBER: 200
MARKS 1 0
MARKS 2 0
MARKS 3 0
MARKS 4 0
MARKS 5 0
AVERAGE: 0
NAME: waleeja
ROLL NUMBER: 101
MARKS 1 34
MARKS 2 56
MARKS 3 67
MARKS 4 78
MARKS 5 90
AVERAGE: 65
PS C:\Sem 2\OOP LABS\lab7> |
```

QUESTION: 3

SOURCE CODE:

1. Header.cpp

```
#include "class.h"
#include <iostream>
using namespace std;

employee::employee(string n, int i, double s){
    emp_name=n;
    emp_id=i;
    basic_salary=s;
}

double employee::calculatebonus() {
    return basic_salary * 0.10;
}

double employee::updatedSalary(){
    return basic_salary+calculatebonus();
}

void employee::displayResult(){
    cout<<"NAME: "<<emp_name<<endl;
    cout<<"EMP ID: "<<emp_id<<endl;
    cout<<"BASIC SALARY: "<<basic_salary<<endl;
    cout<<"BONUS: "<<calculatebonus()<<endl;
    cout<<"UPDATED SALARY: "<<updatedSalary()<<endl;
}
```

2. class.h

```
#include <iostream>
using namespace std;
class employee{
    string emp_name;
    int emp_id;
    double basic_salary;
public:
    employee(string n, int i, double s);
    double calculatebonus();
    double updatedSalary();
    void displayResult();
};
```

3. Main.cpp

```
main.cpp / ...
#include "class.h"
#include <iostream>
using namespace std;

int main() {

    employee emp1("Waleeja", 101, 50000);
    | emp1.displayResult();
    employee emp2("Areej", 202, 60000);
    | emp2.displayResult();

    return 0;
}
```


OUTPUT:

```
PS C:\Sem 2\OOP LABS\Lab7>  
NAME: Waleeja  
EMP ID: 101  
BASIC SALARY: 50000  
BONUS: 5000  
UPDATED SALARY: 55000  
NAME: Areej  
EMP ID: 202  
BASIC SALARY: 60000  
BONUS: 6000  
UPDATED SALARY: 66000  
PS C:\Sem 2\OOP LABS\Lab7>
```